

R.O.B. VISION

VIRTUAL ROBOT COMPANION / MK.01

TECHNICAL REFERENCE

Architecture, frame protocol, virtual model, pairing, and verification

VERSION 0.1.6

UNO Q · RETROPIE · BATOCERA

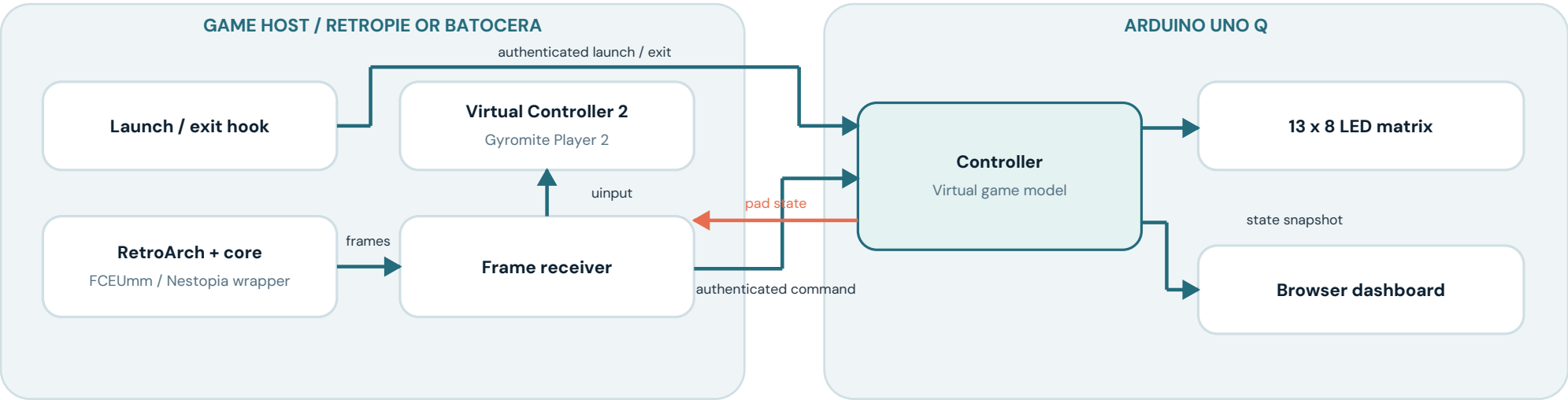


CHAPTER 01 / TECHNICAL REFERENCE

R.O.B. Vision technical reference

This is the implementation reference for the 0.1.x system. The Arduino UNO Q owns Buddy's virtual pose and pieces. RetroPie or supported Batocera runs Gyromite or Stack-Up. A browser on a laptop, tablet, or phone renders the controller's state. Automatic commands come from emulator-rendered NES frames; the current application has no camera input path.

System and responsibilities



Component	Responsibility	Source
Game registry	Exact supported ROM basenames and game IDs	config/games.json, tools/identify_game.py
Console launch hook	Start/end events; never robot motion	RetroPie runcommand hooks or Batocera zz-robvision-game; tools/notify_game.py
Core wrapper	Classify each emulated NES video frame and pass the original video onward	deploy/retropie/rob_vision_fceumm_proxy.c

Component	Responsibility	Source
Console receiver	Check sender, decode commands, report heartbeats, drive virtual Player 2	tools/retropie_frame_hook.py, tools/retropie_controller2.py
UNO Q controller	Game selection, virtual model, pairing, snapshot API	controller/service.py, controller/model.py, controller/pairings.py
UNO Q presentation	Port-80 gateway, LED matrix, browser views	python/main.py, controller/matrix.py, sketch/sketch.ino, dashboard/

The UNO Q is the source of truth for game, pose, pieces, and buttons. The browser polls and animates that state; it does not decode game frames. A directly opened `file:///dashboard` is an independent scripted preview, not the live controller.

Game identity and session lifecycle

The registry matches exact, case-insensitive Gyromite (World) and Stack-Up (World) basenames with `.nes`, `.zip`, or `.7z` extensions. The resolver accepts NES/Famicom metadata, but the installed frame integration and ROM directories are for **NES**. A renamed archive needs a registry entry. The launched archive name matters, not the name inside it. No ROMs or extracted game art ship with the project.

1. A console hook posts a credentialed `start` event with the system and filename to `/api/launch`. The UNO Q resolves it and initializes the matching game's virtual pieces. An unknown title clears the game.
2. The selected R.O.B. Vision wrapper delegates libretro calls to the installed original FCEUmm or Nestopia core and observes video frames. A plain core can play the game but does not provide the frame link.
3. The receiver accepts frames only from an approved wrapper running a registered ROM. It decodes a complete command before posting it to the UNO Q. A launch event alone never moves Buddy.
4. The receiver polls the UNO Q and drives its virtual Controller 2 only while Gyromite is the active RetroArch process. Stack-Up has no controller-return buttons.
5. An end event clears the selected game. The receiver releases the pad. Its RetroArch scan can replay a recognized launch after a UNO Q restart, with retries throttled to two seconds.

The latest authenticated console launch selects the active game and console. While that ownership remains, a different console's frame command is rejected. A manual browser game selection clears console ownership and remains unowned until another authenticated launch. Removing the active console's pairing clears its session. When idle, an online receiver label may refer to either paired console; it is not evidence of a running game.

Frame classification and command protocol

The wrapper intercepts the libretro video callback, classifies the frame, and then forwards the video to RetroArch. It samples a 5-by-5 grid plus nine central picture points. Near-black pixels count as dark; a green-dominant pixel counts as light. A frame is 0 with at least 23 dark grid points, 1 with at least 23 green grid points or eight green central points, and N otherwise. The central sample handles dark borders on Test artwork. These are **emulated frames**, so monitor refresh and camera frame rate are outside the input path.

Each nonblocking Unix datagram is five bytes: a native 32-bit frame index and ASCII 0, 1, or N. It goes to `/run/rob-vision/frames.sock`. The receiver uses Unix process credentials, RetroArch's command line, an approved wrapper path, and exact ROM identity to authenticate the local sender; it rechecks the process at least every half second. A new sender or game resets the decoder.

Both games encode one primitive per 13-frame word, `000101w1x1y1z`, with four variable bits. Stack-Up schedules an extra dark frame after the data word. The [ROM analysis](#) has the examined ROM hashes, CPU addresses, palette proof, and reproducible analysis command.

The inspected Gyromite image has SHA-256 `bf1b323ba39c84b964f93127598b267ff3f16cbf99c533ae1e9b8332232b3e5b`; Stack-Up has `3ab6bc99246783a6bf7083481027fe358ccdb147dcf1165eaf08aaf4e1b06548`. Both are iNES mapper-0 images with 32 KiB program and 8 KiB character ROM. Gyromite's palette-choice table at CPU `$A6FC` supplies 13 entries per populated command slot; its routine at `$A679-$A697` reads them in reverse order. Stack-Up's sender at `$B1DE-$B237` uses the template at `$B24A`. Palette lookups tie the extracted bits to black and green screen frames. These addresses and hashes identify the examined ROM variants, not every regional or patched copy.

Primitive	Gyromite word	Stack-Up word	Model effect
Open	0001011101110	Same	Open hands; release a held piece if placement is valid
Close	0001010111110	Same	Close hands; possibly pick up a piece or block segment
Left	0001010111010	Same	Turn one station left
Right	0001011101010	Same	Turn one station right
Up	0001010111011	0001011111010	Gyromite: two levels; Stack-Up: one
Down	0001011111011	0001010101110	Gyromite: two levels; Stack-Up: one
Ready light	0001011101011	Same	Brief status light; no movement

`ExactFrameDecoder` requires consecutive frame indices and a complete word allowed for the selected game. A gap, N in the candidate, wrong Up/Down code, or incomplete word does nothing. Separate identical complete words are valid. The receiver has a bounded pending queue and briefly retries HTTP delivery; it drops a pending item after one second or when the game/source changes. The UNO Q deduplicates retained retries by sender PID and ending frame index, keeping the last 128 keys. It also rechecks game, pattern, PID, index, and active console. This prevents an ordinary retry from repeating motion; it is not a durable exactly-once guarantee across restarts.

Test detection is separate from command decoding. At least 36 consecutive green frames or 12 alternating edges mark a Test field. A recent authenticated heartbeat pulses the Test light. A complete ready-light word lights it steadily for at most one second. Neither changes pose or pieces. The signal does not reveal the game's menu state through memory.

Virtual model

GET /api/state returns schema 1 with game, robot, events, input, test, and link. The UNO Q keeps a sequence number and the latest 30 display events, not a durable event log. Browsers poll every 500 ms. A refresh reads the current snapshot. A fresh input.frame_hook means matching game frames recently reached the receiver; it does not prove that the game gate moved or a Stack-Up objective was completed.

Gyromite

The model tracks two gyros, six height levels, grip, held gyro, spin deadlines, and manual Gate Assist deadlines across five ordered stations: holder B, holder A, red pad, blue pad, spinner. Left/Right moves one station; Up/Down moves two levels. Grasp and release require valid station and height; turning with a low held gyro is blocked. Placing a gyro on the spinner starts an illustrative 55-second spin clock. A spinning gyro on a colored pad presses its button. A held gyro over a pad at low height may press it without spinning. The controller does not measure physical spin or inspect gate pixels.

Fast Gates is a manual browser control. It temporarily puts an available gyro on a pad and holds that button for at most 60 seconds, then returns the gyro to its holder on release or timeout. Game-frame movement is rejected while Gate Assist is held. Red and blue are independent. Selecting a game or Home rebuilds the pieces; Emergency Stop selects no game.

Stack-Up

The model tracks five trays, five distinct blocks, six height levels, one station, and one grip. It starts all blocks on Tray 3, bottom to top green, yellow, blue, white, red. Left/Right moves one tray; Up/Down moves one level. Closing at a block height lifts that block and everything above it as an ordered segment. A carried segment must clear another tray; release requires the next free level. Invalid moves leave pieces in place, and a conservation check rejects loss or duplication.

Direct, Memory, and Bingo transmit the same six movement primitives. The controller does not receive target pattern, score, or victory state. It currently uses one starting arrangement; distinct mode-specific historical layouts and extended Memory/Bingo play are outside the verified model claim.

Gyromite return path

The console receiver creates a Linux /dev/uinput pad named R.O.B. Vision Controller 2. It polls the UNO Q every 50 ms with a 250 ms request timeout, validates schema 1, and emits changed red/blue states with a sync event. It releases both when Gyromite is not the active RetroArch process, the paired source is not selected, or no good response arrives for 750 ms. Exit and service shutdown also release them.

Runtime setting	Current value	Owner
Browser snapshot poll	500 ms	dashboard/live.js
Receiver poll / HTTP timeout / stale pad release	50 / 250 / 750 ms	tools/retropie_controller2.py
RetroArch process scan / launch replay throttle	250 ms / 2 s	tools/retropie_controller2.py
Frame-link freshness / receiver-online freshness	1 s / 3 s	controller/service.py

Runtime setting	Current value	Owner
Recent event history / retry keys	30 events / 128 keys	controller/service.py
Gyro spin / manual gate maximum	55 / 60 s	controller/model.py
Game registry	config/games.json	Console and UNO Q
Frame socket	/run/rob-vision/frames.sock	Console receiver

RetroPie installs a virtual-pad RetroArch profile, detects the current joystick index, and configures NES Player 2 as a gamepad. Batocera detects joystick order through SDL2, which may differ from Linux /dev/input/jsN. It writes persistent Player 1 index overrides for registered games and **Gyromite-only** Player 2 overrides before Batocera generates RetroArch's runtime config. Its receiver swaps Linux buttons to match the Batocera profile. The physical controller remains Player 1; Buddy's pad is Player 2. Nestopia interprets device 1 as Auto, so its wrapper selects explicit gamepad device 257 after ROM load. The tested FCEUmm and Nestopia paths returned Gyromite's blue gate after release; the [release validation](#) records observed results.

Pairing, API, and trust boundary

App Lab starts a port-80 gateway that forwards to controller/service.py on port 8766. The UNO Q's existing hostname and mDNS service provide its .local address; R.O.B. Vision does not advertise a separate name. A reachable LAN IP works too. Wi-Fi and Ethernet carry the same HTTP protocol. Keep ports 80 and 8766 on a trusted LAN.

Endpoint	Effect	Access
GET /api/state	Browser snapshot; receiver poll carries frame/Test headers	LAN read; credential verified when supplied
GET /api/matrix/state	Matrix bridge status	LAN read
POST /api/launch	Console start/end	Paired console credential
POST /api/emulator/command	One ROM-derived word	Paired credential and active-source checks
POST /api/game, /api/command, /api/gate-assist	Manual browser controls	Trusted-LAN browser, JSON and same-origin checks
POST /api/pair, /api/consoles/remove	Pair or revoke console	Trusted-LAN browser; pairing also checks code and certificate fingerprint
POST /api/matrix/pairing	Show or clear pairing cue	Trusted-LAN browser

On first install, a temporary TLS pairing server on console port 8768 prints a six-digit code and SHA-256 certificate fingerprint. Its window is five minutes, with a limit on wrong attempts. Setup sends a newly generated console ID and bearer credential only after checking the fingerprint and private-network destination. The console stores them with owner-only permissions; the UNO Q stores a separate revocable record per console. An upgrade retains that credential. Removing a console revokes the UNO Q record; `--pair` establishes a fresh one. After pairing, console traffic uses authenticated LAN HTTP. Browser controls intentionally have no token on the trusted LAN. JSON, Fetch Metadata, and Origin checks reduce cross-site writes but do not replace network isolation.

Matrix and browser

The sketch owns the 13-by-8 LED framebuffer and starts an hourglass before the Linux bridge is ready. `controller/matrix.py` sends a mode and occasional action hint through Router Bridge. Priority is pairing, Test/ready, selected game, then idle eyes. Games show GY or SU followed by eyes; Test pulses T; pairing pulses P. A fresh movement briefly changes the eyes. Linux refreshes at least once per second. A bridge heartbeat gap of about 3.5 seconds returns the sketch to the hourglass. The sketch contains an X glyph, but the controller does not currently request its fault mode. The [Matrix Display Guide](#) explains the active cues to players.

Mission animates the model, accessory views, Game Table, Pose Preview, vitals, and recent activity from the same snapshot. Setup handles pairing, link checks, Test status, and manual game checks. A paired console may be online while no game is selected. Browser demo runs while paired and idle; a live game takes authority. The UI cannot establish Hector's location, Stack-Up scoring, or full-game completion.

Installation and platform integration

The versioned release installer downloads a source package, checks its embedded SHA-256 before extraction, then dispatches to UNO Q or console installation. Release assets also contain checksums and PDFs. The UNO Q installer stages its App Lab app, stops and restarts it, and preserves pairing data. App Lab compiles and uploads the matrix sketch. The installer prints the device's `.local` and available LAN IPv4 dashboard URLs. On one UNO Q, only one App Lab app can own port 80 at a time. Separate UNO Q devices can each serve their own application and hostname.

RetroPie installs launch/end hooks, a systemd receiver, a virtual-pad RetroArch profile, and FCEUmm/Nestopia wrapper launch choices. Its installer builds wrappers against installed cores and detects the Player 2 joystick after first pairing. It retains an existing per-ROM supported core choice rather than forcing every host to FCEUmm. EmulationStation must be closed while the virtual input service is replaced.

Batocera 43.1 x86_64 uses packaged wrappers, a separate R0BVision service and game hook, runtime core overlay, and persistent per-ROM overrides. The installer preserves an existing supported Nestopia choice; otherwise it selects the FCEUmm wrapper for registered ROMs. Other board/version combinations are rejected before writes. It suspends and resumes EmulationStation around receiver restart. When VirtualGlove is installed and enabled, R0BVision waits for its core overlay before adding its own wrappers; it does not stop or disable VirtualGlove. The [installation guide](#) has the supported command and prerequisites; the [configuration reference](#) lists tunable defaults.

Recovery and verification boundary

Condition	Behavior
Browser refresh	Reads UNO Q snapshot without resetting
Unknown title or game exit	Clears selected game; virtual buttons release
UNO Q restart during a registered game	Receiver can replay launch; frame link resumes after authenticated polling and wrapper frames
Missing, non-light, or malformed frame	Drops partial word; no repeated motion
No good receiver response for 750 ms	Both Gyromite buttons release
No authenticated receiver poll for three seconds	Console status becomes offline
No matching game frames for one second	<code>input.frame_hook</code> becomes false
No matrix heartbeat for about 3.5 seconds	Sketch returns to hourglass

Operational checks

Observation	Interpretation	Next check
Game selected, frames waiting	Launch hook reached UNO Q; the approved wrapper has not produced fresh matching frames	Confirm the registered ROM launched with the R.O.B. Vision FCEUmm or Nestopia choice
Game frames linked, no movement	Fresh frames arrived; no valid command may have completed, or the model may have blocked one	Send a Direct-mode command and read the latest <code>decoded</code> , <code>action</code> , or <code>blocked</code> event
Console online, game idle	Receiver polling works; no recognized active game	Check exact ROM basename and console launch hook
Gyromite action visible in Mission, gate unchanged	Model changed, but the return path or game state may be wrong	Check Player 2 mapping, active RetroArch process, and independent blue/red holds
Test light pulses, pose unchanged	Test field recognized as status	Leave Test mode and confirm the light clears before a Direct command
Matrix hourglass while dashboard is reachable	Linux service may be up while Router Bridge or sketch heartbeat is unavailable	Read <code>/api/matrix/state</code> and check App Lab/sketch state

The checks support different claims: a recognized ROM proves selection; a fresh frame link proves source delivery; a decoded event proves a complete word; an `action` event proves the virtual model accepted it; a visible gate response proves Controller 2 returned input to Gyromite. Do not collapse these into one "connected" result.

Verified on supported test hosts	Still outside the demonstrated claim
Real launch and exit detection for both games	Unknown-title interactive launch on every host
All six Direct-mode movement commands from both games under FCEUmm	Long unattended sessions and measured missed-command rate
Movement commands under Nestopia for both games	Full six-command Nestopia sessions under every game mode
Gyromite blue-gate response and return under both cores	Full-game gate automation and game outcomes
Batocera 43.1 x86_64 reboot with VirtualGlove coexistence	Other Batocera board/version combinations
Automated model, decoder, installer, and pairing tests	Stack-Up Memory/Bingo behavior and complete matrix visual checks

A valid frame link alone does not imply model success: an accepted command can be blocked at a boundary. The browser does not know Hector's position or the Stack-Up goal. The matrix's active modes are implemented, while complete direct visual checks remain open.

The [verification plan](#) tracks open checks. The [FCEUmm report](#) and [Nestopia report](#) preserve dated evidence. The [engineering journey](#) explains how camera testing led to the frame link. Earlier [camera row tests](#) and [optical field notes](#) are historical research and do not describe the current input path.